INCH-POUND
A-A-50199C
April 28, 2025
SUPERSEDING
A-A-50199B
July 31, 2018

COMMERCIAL ITEM DESCRIPTION

THREAD, POLYESTER CORE, COTTON OR POLYESTER-COVERED

The General Services Administration has authorized the use of this Commercial Item Description (CID) by all federal agencies.

1. SCOPE. This CID covers the requirements for a polyester core, cotton or polyester- covered thread. The commercial thread shall be used for machine sewing of clothing and other light-medium weight textile assembly applications.

2. CLASSIFICATION. This commercial thread shall be of the following types:

2.1 Types:

Type I – Cotton-covered
Type II – Polyester-covered

3. SALIENT CHARACTERISTICS.

3.1 General description. Each ply of the thread shall consist of a continuous, multifilament polyester end, spun with one (1) or more ends of cotton, or high tenacity staple polyester roving to cover the multifilament polyester end. The direction of the final twist shall be "Z", unless otherwise specified. The thread shall be as specified in Table I and II.

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System ASSIST online database https://assist.dla.mil.

AMSC N/A FSC 8310

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3.2 Physical characteristics requirements. The Type I and Type II thread shall conform to the physical characteristics specified in Table I when tested as specified in Table II.

TABLE I. Physical characteristics (all types).

Tex 1/	Ply	Breaking strength (lbs.), minimum		Elongation (%), maximum
		Type I	Type II	
21-23	2	1.4	1.8	30
24-29	2	2.0	2.4	30
30-34	2	2.6	3.0	30
35-44	2	3.2	3.7	30
45-59	2 or 3	4.0	4.7	30
60-69	2 or 3	5.0	6.1	32
70-104	2, 3 or 4	6.8	7.2	32
105-149	2, 3 or 4	9.9	10.8	32
150-209	2, 3 or 4	13.0	14.1	32

1/ Tex size based on weight in grams/1,000 meters thread.

TABLE II. Physical requirements and test methods.

Characteristic	Requirement	Test method
Fiber identification	3.1	AATCC TM20 (see 7.6)
Direction of twist	3.1	ASTM D204
Tex	Table I	ASTM D204 and ASTM D3823
Ply	Table I	Visual
Breaking strength (lb.f),(minimum)	Table I	ASTM D204 1/
Elongation (%), (maximum)	Table I	ASTM D204 1/
Colorfastness (minimum):		
Laundering after three (3) cycles	3-4	AATCC TM61, Test 3A, 2/, 3/
Light (after 40 AFU or 170kJ/(m^2nm)@420nm)	3-4	AATCC TM16.3 2/,
Perspiration	3-4	AATCC TM15, 2/, 4/
Dry heat @ 376 (±) 6°F	3-4	AATCC TM117, 2/
Weathering (after 40kJ/(m^2nm)@420 nm)	3-4	AATCC TM169, 2/
Visual shade matching	3.3	3.3.1
Non-wicking finish	3.5.1	3.5.1.1
Toxicity	3.6	3.6.1

1/ Five (5) determinations shall be made on the sample unit.
2/ AATCC EP1, Gray Scale for Color Change.
3/ The color transfer cloth evaluation shall not apply.
4/ AATCC EP2, Gray Scale for Staining.

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3.3 Color. The color shall be as specified in the applicable end item specification or in the contract or purchase order. The dyed thread shall be a good match to the applicable end item when examined in accordance with 3.3.1. The shade of the polyester core shall approximate that of the cotton or staple polyester covering.

3.3.1 Visual shade matching. The color and appearance of the finished thread shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 7.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the finished thread shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 7.5.1). The finished thread shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

3.4. Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The cloth shall be "Free" of labile sulfur when tested as specified in 3.4.1.

3.4.1 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin (II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35 percent hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one (1) at a time) of glacial acetic acid until the solution is clear. The samples shall be submerged in the acidic tin (II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin (II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as "pass" or "fail"

3.5 Finishing materials. The finished thread shall have no chemical finishes or treatments other than those commonly used (e.g. lubricants, etc.) on commercial threads or as specified in the contract (see 3.5.1) which have been demonstrated to have no harmful effects on the fiber, including effects of prolonged storage. No finish or treatment shall be applied for the purpose of increasing breaking strength. There shall be no noticeable wicking of the treatment on the thread to adjacent material when sewn.

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3.5.1 Non-wicking finish. When specified, the thread shall have a uniformly applied commercial non-wicking finish. The use of compounds containing mercury in any form shall not be used. The finished thread shall resist the wicking of water when tested as specified in 3.5.1.1.

3.5.1.1 Vertical resistance to wicking. The thread shall be water repellent treated so that the treated thread shall resist the wicking of water for a period of not less than 2-hours when tested as follows: The test specimen shall consist of a 20-strand skein of thread in one (1) continuous 30-yard length made on a 54-inch periphery skein reel. The skein shall be reeled under enough tension to cause the strands in the skein to lie uniformly, side by side, on the reel. The finishing end of the skein shall be tied to the starting end of the skein in such a manner that the knot will not add additional length to the reel skein. The skein shall be hung over the movable crossbar of a laboratory stand with the end hanging over the vessel. The movable crossbar shall rise 28-inches or more above the base. A non-ferrous 3/4 to 7/8 ounce weight shall be placed in the lower catenary of the skein to keep it taut and straight. The skein shall be arranged so that the strands are touching each other in flat ribbon form. The vessel shall be filled to a depth of at least 5-inches with distilled water at a room temperature which has been mixed with 0.05 percent blue food coloring (salt and wetting agent free). A piece of blotting paper shall be attached by means of a paper clip or similar clamp to one (1) full side (20-strands) of the skein, 3-inches above the lower catenary of the skein. The position of the crossbar shall be adjusted that when the skein is hung freely in the liquid, 2-inches of the skein will be immersed in the liquid and the lower edge of the blotter is 1-inch above the liquid surface. The skein shall then be slowly lowered into the dyebath and the time of entry shall be noted. Depending on the dimensions of the vessel and the length of the crossbar, several specimens can be tested at the same time in the same dyebath by hanging the skeins sufficiently apart on the crossbar. The skein shall be exposed for 2-hours. The blotter shall be examined for wetting or staining at least every hour. The test shall be terminated whenever staining or wetting of the blotter is observed within the 2-hour duration. Staining or wetting before the 2-hour time frame shall constitute a failure.

3.6 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished thread shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 3.6.1.

3.6.1 Toxicity test. When the toxicity requirement (see 3.6) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 7.1.4 and 7.6).

3.7 Put-up. Unless otherwise specified, the thread shall be put-up on holders such as commercial spools, cones, tubes, or bobbins as specified in the contract. The thread shall be wound around the specified holder in one continuous piece, so that each turn and layer is free of entanglement. The outside ending of the thread shall be secured to prevent unwinding, loosening, or slippage during handling, shipping, or storage.

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3.8 Labeling. Each thread holder shall have a label, adhered securely to remain in place and be clearly legible until all thread has been removed. The label shall be printed and include information related to length in yards or weight of cone, direction of twist, color, Tex size, name of thread manufacturer, and nomenclature specifying fiber Type and construction.

3.9 Workmanship. The finished thread shall conform to the quality of the product established in this document. The thread shall average not more than one full thread knot or splice per 1,000 yards. The thread shall be free from defects which would affect its serviceability, appearance, uniformity and quality assurance standards defined by the technical data in the procurement documents.

4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation FAR, or the sustainable acquisition of EO 13693; as applicable. (see 7.1.2).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 End item examination. The visual examination for defects shall be in accordance with ASQ/ANSI Z1.4 (see 7.2.2).

5.3 Visual examination. Thread shall be examined for the defects listed in Table III.

TABLE III. Visual examination defects.

Knots: Tex 45-49 and finer thread not more than one (1) thread knot per 8-ounces in singles.
Color: Not as specified.
Labels: Label missing, incorrect, or illegible. Required information missing from the label.
Workmanship: Average full thread knot or splice more than specified. Failure to meet product conformance; serviceability reliability and quality as defined by this document.
Packaging: Not packaged in accordance with the contract or purchase order

5.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.8).

6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see 7.8).

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7. NOTES

7.1 Sources of Government documents.

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil

7.1.2 EO 13693 information is available online at https://www.epa.gov.

7.1.3 Federal Acquisition Regulations (FAR) are available online at https://acquisition.gov.

7.1.4 Toxicity testing. If dermal toxicity testing is required, it will be conducted in accordance with Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1).

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100).
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250).

7.2 Sources for non-Government documents.

7.2.1 AATCC test methods are available online at https://aatcc.org.

7.2.2 ASQ/ANSI Z1.4 - Sampling Procedures are available online at https://asq.org.
Or https://ansi.org.

7.2.3 ASTM Standards are available online at https://www.astm.org.

7.3 Intended use. The threads are intended for machine sewing of clothing and other light to medium weight textile assembly applications. Treated threads with a non-wicking finish can also be used on items where water resistance of seams is required.

7.4 Standard samples. For access to standard shade samples of thread, address the contracting activity issuing the invitation for bids or request for proposal.

7.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C.
NOTE: In case of confusion, the viewing geometry should be “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens”.
7.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222’s should reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

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7.6 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

7.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issues due to the extensiveness of the changes.

7.8 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this Commercial Item Description (CID).
b. Type and Tex size required (see 2.1, 3.2).
c. Color required (see 3.3).
d. When non-wicking finish is required (3.5.1).
e. When toxicity testing is required (see 3.6 and 3.6.1).
f. Put-up required if other than specified (see 3.7).
g. Product conformance (see 5.1).
h. Acceptance criteria (see 5.4).
i. Packaging requirement (see 6.).

7.9 Key words.

Clothing
Machine
Non-wicking
Sewing
Textiles

MILITARY INTERESTS:

Custodian:
Army-GL
Navy- NU
Air Force- 11

Review Activities:
Army- MD
Navy- AS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing activity
DLA –CT
:
(Project: 8310-2025-001)

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